

JAMES GUILLOCHON

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Publications: **125** (26 first author w/ 2,529 citations, h-index 14)

Education

Harvard — Postdoctoral Scholar, Astronomy — Cambridge, MA	June 2018
Insight Data Science — Fellow, Data Science — Boston, MA	June 2018
UC Santa Cruz — PhD, Astronomy — Santa Cruz, CA	June 2013
UC Irvine — BS, Physics — Irvine, CA	June 2005

Work Experience

Esri — Principal Data Scientist, San Diego, CA Jan 2021 - Present (5 years)

- **Generative AI & LLM Tooling:** Contributed to generative AI workflows by implementing tools that integrate large language models for image interrogation, free-form text object detection, and automated object replacement.
- **Advanced Imagery & Zero-Shot Detection:** Spearheaded advanced imagery analysis initiatives utilizing zero-shot detection and conducted comprehensive statistical evaluations on the resulting AI outputs.
- **3D Scene & LIDAR Optimization:** Optimized 3D LIDAR processing pipelines and applied PointCNN models alongside advanced AI algorithms and image embeddings to identify infrastructural assets within 3D scenes.
- **Oceanic Predictive Modeling:** Developed deep many-channel CNN models and forecasting systems to predict the worldwide occurrence of rogue and anomalously large waves based on current oceanic conditions.
- **Environmental Deep Learning:** Created automated deep learning and hyperspectral models to process satellite and sensor data for accurate land classification and the prediction of harmful algal blooms.
- **Aerial Feature Extraction:** Applied transfer learning techniques to aerial imagery to generate precise, pixel-level labels for widespread infrastructural features, such as unpaved road networks.
- **Time-Series Imagery Monitoring:** Developed proof-of-concept pipelines to retrieve and analyze time-series satellite imagery for monitoring culturally significant sites impacted by regional conflicts.
- **Automated Anomaly Detection:** Designed and deployed automated solutions to actively monitor open-source map catalogs and identify suspicious changesets.
- **NLP & Document Parsing:** Authored parsing tools to analyze scanned legal documents and automatically extract and map the geographic regions described within the text.
- **Leadership & External Enablement:** Directed analyst teams to generate high-quality satellite imagery datasets and delivered technical workshops and presentations on AI enablement to external organizations.

Berkshire Grey — Principal Data Scientist & Robotics Engineer, Lexington, MA Oct 2018 - Dec 2020 (2 years)

- **Robotics Software Stack:** Led optimization efforts of Python/ROS software stack used for pick and place in a multi-component robotics system, including robotic arms, high-performance conveyors, and automated shuttles.
- **Motion Planning & Parameter Prediction:** Created and integrated parameter prediction to select optimal robotic motions for unseen products on demand.
- **Bayesian Sensing & Pick Verification:** Developed online Bayesian analysis of data from sensors integrated into the picking system to dynamically determine how many items the robotic arm lifted.
- **Operations Analytics:** Curated a dashboard of key performance indicators (KPIs) for the deployed robotic system via the ELK Stack.

Harvard-Smithsonian — NASA Einstein Fellow & ITC Fellow, Cambridge, MA Sept 2013 - Aug 2018 (5 years)

- **Research Leadership:** Technical lead of an eight-person research team of undergraduate/graduate students and postdocs resulting in [dozens of completed scientific projects](#) with 4k+ citations.
- **Science Communication:** Appeared on the [NOVA television program](#) in 2018, describing data analysis by my group to measure a black hole's mass using a recently destroyed star.
- **MOSFiT & Bayesian Modeling:** Developed [MOSFiT](#), a Bayesian engine to infer the most likely analytical models for time-series data.
- **Open Astronomy Catalogs:** Led creation of the [Open Astronomy Catalogs](#), a platform for sharing astronomical data, with a user-facing frontend and an API backend to serve data quickly to users.